Prediction of Chronic Disease Risk Using Nutrition, Lifestyle & Clinical Data from The Irish Longitudinal Study on Ageing (TILDA)

UNIVERSITY OF LIMERICK
OLLSCOIL LUIMNIGH

Abderahman Haouit

CSIS Research Centre

Department of Computer Science and Information Systems

Faculty of Science and Engineering

University of Limerick

Submitted to the University of Limerick for the degree of

Artificial Intelligence (MSc) 2026

1. Supervisor: Dr. Alison O'Connor
Computer Science & Information Systems
University *of* Limerick
Ireland

Abstract

Background: Chronic diseases such as Cardiovascular Disease (CVD), Diabetes Mellitus (DM), Frailty, Gait, Exercise, Falls and Fractures (FR), and obesity represent major contributors to morbidity in older adults. While traditional prediction models focus heavily on clinical and functional indicators, the contribution of nutritional and lifestyle factors to early risk detection remains underexplored.

Methods: This thesis applies machine learning techniques to data from The Irish Longitudinal Study on Ageing (TILDA), a national representative longitudinal study of adults aged 50+. Five waves of socio demographic, clinical, nutritional, psychological, and behavioural data, along with the externally produced Research and Development Corporation (RAND) Harmonised The Irish Longitudinal Study on Ageing (TILDA) dataset, were systematically cleaned and standardised using a multi stage preprocessing pipeline. Predictive models, including (*mention models used*) were developed to estimate the risk of chronic disease and Frailty, Gait, Exercise, Falls and Fractures (FR) related outcomes. Model performance was evaluated using (*mention examples*), and feature importance was used to assess predictive value.

Results: TBD

Conclusions: TBD

Keywords: ageing, (Artificial Intelligence (AI)), Machine Learning (ML), predictive modelling, Frailty, Gait, Exercise, Falls and Fractures (FR), chronic disease, nutrition, The Irish Longitudinal Study on Ageing (TILDA)

Declaration

I Abderahman Haouit declare that I have produced this thesis without the prohibited assistance of third parties and without making use of aids other than those specified; notions taken over directly or indirectly from other sources have been identified as such. This thesis has not previously been presented in identical or similar form to any other Irish or foreign examination board.

The thesis work was conducted from 2025 to 2026 under the supervision of Dr. Alison O'Connor at University of Limerick.

Limerick, 2026

Acknowledgements

I would like to acknowledge my supervisor, Dr. Alison O'Connor, for her guidance, support, and for providing a well organised environment throughout the implementation of this project.

I would also like to thank the UL Science and Engineering Research Ethics Committee for approving the ethical aspects of this research. Special thanks to the ISSDA team for facilitating access to the TILDA dataset, collected by Trinity College Dublin.

Your dedication goes here. Open the dedication.tex file found at:
0_frontmatter/dedication.tex

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List of Abbreviations

AI	Artificial Intelligence
AUC	Area Under the ROC Curve
AUROC	Area Under the Receiver Operating Characteristic Curve
BMI	Body Mass Index
BP	Blood Pressure from Sphygmomanometer
brainPAD	Brain predicted age difference
CAPI	Computer Assisted Personal Interview
CES-D	Center for Epidemiologic Studies Depression Scale
COG	Scales and Questions on Cognition
CRT	Choice Reaction Time (Cognitive Test)
CV	Cross Validation
CVD	Cardiovascular Disease
DIS	Disability, Functional Impairment and Helpers
DM	Diabetes Mellitus
DNN	Deep Neural Network
EU	European Union
F1	F1-score (harmonic mean of precision and recall)
FOF	Fear of Falling
FR	Frailty, Gait, Exercise, Falls and Fractures
GBM	Gradient Boosting Machine
GDPR	General Data Protection Regulation
HADS	Hospital Anxiety and Depression Scale
HGS	Hand Grip Strength

HRS Health and Retirement Study, (a longitudinal study of ageing in the United States)

IADL Instrumental Activities of Daily Living
INC Income Variables
IPAQ International Physical Activity Questionnaire
ISSDA Irish Social Science Data Archive

MD Medication Data
MGS Maximum Gait Speed
MH Scales and Questions on Mental Health and Wellbeing
ML Machine Learning
MVPA Moderate-to-Vigorous Physical Activity

NOS Newcastle-Ottawa Scale

OR Odds Ratio

PA Physical Activity
PMF Pseudonymised Microdata File

RAND Research and Development Corporation
Recall Recall (true positive rate)
RF Random Forest

SCQ Self Completion Questionnaire
SCR Screening Tests
SHAP Shapley Additive Explanations
SLR Systematic Literature Review
SOC Social Variables
SR Systematic Review

TILDA The Irish Longitudinal Study on Ageing
TUG Timed Up and Go (mobility test)

UGS Usual Gait Speed

WHO World Health Organization

List of Abbreviations

Chapter 1

Introduction

Ageing populations experience substantial changes across physical, nutritional, cognitive, and psychosocial domains, each shaping vulnerability to chronic disease and functional decline [4–6]. Understanding how these multidomain factors interact over time is essential for early risk identification.

The financial and societal burden is substantial. In Europe, poor diet is a major risk factor for non communicable diseases, contributing billions in healthcare costs annually [7]. In Ireland, malnutrition alone is estimated to cost over €1.4 billion per year [8]. Recognising this, international and national policies including the World Health Organization (WHO) Global Action Plan on Noncommunicable Diseases and Ireland's *Healthy Ireland* framework highlights nutrition and lifestyle interventions as central to healthcare strategy.

European regulatory frameworks such as the General Data Protection Regulation (GDPR) and proposed European Union (EU) Artificial Intelligence (AI) Act emphasise transparency and explainability, highlighting the need for interpretable Machine Learning (ML) models in healthcare.

Traditional risk prediction models have largely relied on clinical biomarkers such as Blood Pressure from Sphygmomanometer (BP) and lipid profiles. Yet, growing evidence shows that broader predictors including nutrition [9], Physical Activity (PA) [10], Scales and Questions on Mental Health and Wellbeing (MH) [11], and Social Variables (SOC) [12] improve predictive performance. Estimating the likelihood of developing a condition given certain risk factors helps quantify the influence of these predictors in clinical decision making.

The Irish Longitudinal Study on Ageing (TILDA)[1] provides a national representative cohort of adults aged 50+ in Ireland, surveyed repeatedly across multiple domains. For benchmarking and international comparison, the externally produced Research and Development Corporation (RAND) Harmonised TILDA dataset (available for Waves 1 and 2) aligns selected variables with the Health and Retirement Study, (a longitudinal study of ageing in the United States) (HRS) framework. Full preprocessing and alignment details are mentioned in Chapter 3.

Despite these advantages, existing ML research on chronic disease prediction faces persistent limitations: over reliance on clinical predictors, limited incorporation of lifestyle and SOC variables, predominantly cross sectional study designs that fail to capture trajectories, and a lack of interpretability consistent with emerging AI governance standards.

This thesis addresses these gaps by leveraging longitudinal TILDA's data across multiple domains, using interpretable ML methods to combine modifiable health behaviours, psychosocial factors, and established clinical measures to enhance prediction of ageing-related outcomes.

1.1 Research problem and motivation

Despite increasing policy interest in lifestyle and nutrition and advances in ML methods, approaches to chronic disease prediction in older adults remain limited in several ways.

- **Limited Integration of Multidomain Predictors:** Existing models focus primarily on clinical measurements and often neglect modifiable behavioural, dietary, and psychosocial factors [9, 13].

- **Underuse of Longitudinal Dynamics:** Few studies leverage repeated measures to capture developments of risk, such as Frailty, Gait, Exercise, Falls and Fractures (FR) progression [14]. Probabilistic reasoning can help estimate the likelihood of developing a condition at a future time given earlier measurements, for example calculating $P(\text{FR at wave 5} \mid \text{UGS at wave 1}, \text{Nutrition at wave 2})$

[1] TILDA data are pseudonymised and accessed under license via Irish Social Science Data Archive (ISSDA). Use adheres to UL S&E ethics approval and data governance requirements.

This approach helps quantify how early indicators influence later outcomes and supports better feature selection and timing of interventions in predictive models.

- **Lack of Explainability:**Many ML models are not interpretable, limiting clinical adoption and failing to meet emerging regulatory requirements for trustworthy AI in healthcare [15].

Addressing these challenges is essential to improve predictive accuracy, support evidence based interventions, and ensure alignment with evolving policy and regulatory priorities.

1.2 Research aims and objectives

The goal of this thesis is to develop and validate ML models that integrate nutrition, lifestyle, and MH features with clinical data to improve prediction of chronic disease and ageing related outcomes in a representative cohort of older adults. This goal will be achieved through the following objectives:

- Review existing applications of ML for chronic disease prediction in ageing populations, with emphasis on gaps in lifestyle and nutrition integration.
- Develop a reproducible pipeline to preprocess and align TILDA waves 1–5, and engineer multidomain features (nutrition, activity, mobility, MH, clinical indicators).
- Develop and compare baseline statistical models (e.g., logistic regression) with advanced interpretable ML methods Random Forest (RF), Gradient Boosting Machine (GBM).
- Evaluate models using discrimination Area Under the ROC Curve (AUC), calibration, and error metrics (accuracy, precision, Recall (true positive rate) (Recall), F1-score (harmonic mean of precision and recall) (F1)), perform ablation analyses to assess the incremental value of lifestyle and nutrition predictors.

- Identify actionable predictors (e.g., vitamin D status, UGS reserve) and highlight modifiable risk factors for intervention.
- Document ethical, governance, and reproducibility practices for handling longitudinal pseudonymised health data.

1.3 Contributions of the research

This research makes contributions at three levels:

- **Empirical:** Demonstrates how behavioural, dietary, and MH features complement established clinical measures to enhance predictive performance.
- **Methodological:** Provides a reproducible pipeline for aligning longitudinal TILDA data across domains and integrating diverse features into ML workflows.
- **Clinical:** Highlights actionable modifiable factors alongside traditional clinical indicators for targeted interventions and policy.

1.4 Thesis Outline

In *Chapter* 2 a literature review on ML prediction of chronic disease in ageing populations, with emphasis on lifestyle and nutrition predictors and methodological considerations is presented. *Chapter* 3 discusses the TILDA study design, data preprocessing, alignment, and feature engineering (Flow Charts...). In *Chapter* 4 the modelling framework is outlined. This includes baseline and advanced ML, hyperparameter tuning, validation, and evaluation metrics. *Chapter 5* shows the model result, including model comparisons, assessments of feature importance, and an ablation analyses. In *Chapter 6* the implications, limitations, ethical considerations are discussed, and future directions for the research proposed. Additional supporting documents are included in appendices.

Chapter 2

Literature Review

This chapter reviews literature on ageing, chronic disease, and predictive modelling using TILDA (introduced in Chapter 1). It focuses on three interrelated domains central to later life health: FR and physical function, MH, and nutrition within the context of biological ageing. This review focuses on FR, MH, and nutrition as key determinants of later life outcomes. In line with the WHO healthy ageing framework, research using TILDA highlights FR, MH, and nutrition as interconnected determinants of later life outcomes. The following subsections examine each domain in turn.

2.1 Methodology

Several sources describe how systematic and integrative reviews are commonly carried out in fields such as software engineering, health, and education. The approach followed key principles for systematic mapping, for example defining research questions, identifying relevant studies through structured database searches, applying inclusion and exclusion criteria, and categorising results as outlined in [1]. The integrative literature review [16] helped structure and connect findings across studies, while [17, 18] guided the organisation of ideas. In related methodological work, [2] clarified how Systematic Review (SR)s can identify research gaps, and [19] showed how mapping and synthesis highlight trends and future directions. This combined guidance informed a structured yet flexible methodology suitable for ageing, AI, and health research.

2.1.1 Search strategy and database screening

The main topic of interest was chronic disease prediction using nutrition, lifestyle, and clinical data from TILDA. Relevant databases (e.g., ScienceDirect, PubMed) were searched using combinations of keywords such as “Irish Longitudinal Study on Ageing”, “machine learning”, “chronic disease”, “nutrition”, “diet”, and “physical activity”.

The search covered studies published from 2011 to 2025. Figure 2.1 summarises the number of studies identified per year, highlighting trends and gaps over time.

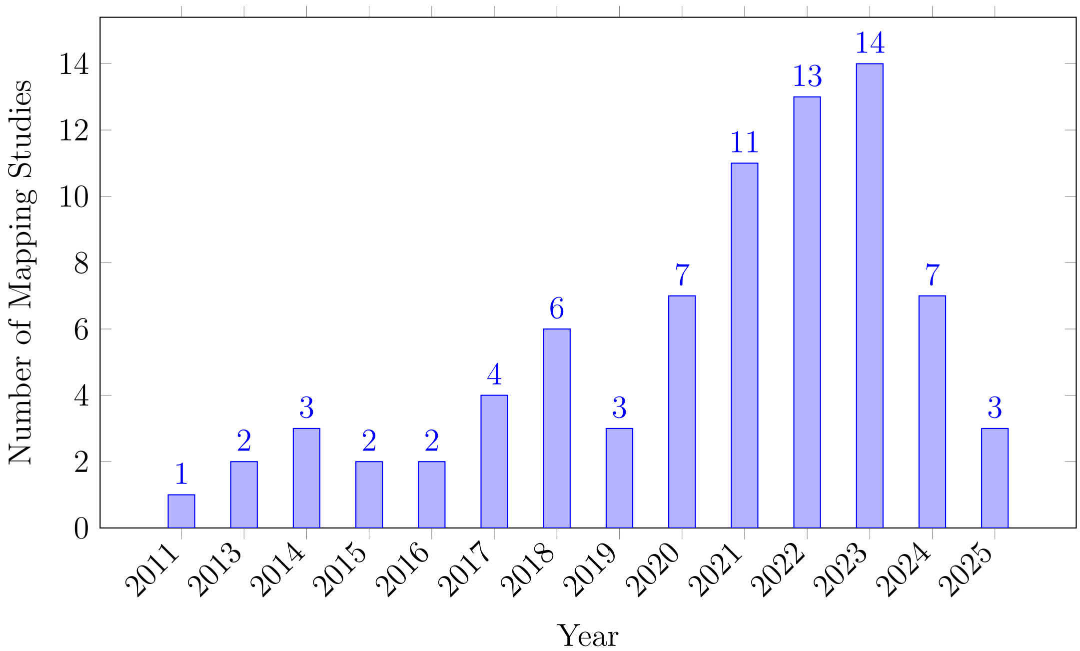

Figure 2.1: Publications per year.

Table 2.1: Database searches and number of studies per database 2.3 [1].

Database	**Search**	**Results**
Oxford Academic	(TILDA OR Irish Longitudinal Study on Ageing)AND (chronic disease OR diabetes OR cardiovascular) AND (nutrition OR diet OR "physical activity")	1081
	(TILDA OR "Irish Longitudinal Study on Ageing") AND ("machine learning" OR "predictive model" OR "risk prediction")	71
IEEE Xplore	TILDA OR "Irish Longitudinal Study on Ageing"	56
	(TILDA OR "Irish Longitudinal Study on Ageing") AND ("machine learning" OR "predictive model" OR "risk prediction")	7
ScienceDirect	(TILDA OR "Irish Longitudinal Study on Ageing") AND ("chronic disease" OR cardiovascular OR diabetes) AND (nutrition OR diet OR lifestyle OR "physical activity")	257
	(TILDA OR "Irish Longitudinal Study on Ageing") AND ("machine learning" OR "predictive model") AND ("chronic disease" OR diabetes) AND (nutrition OR "physical activity")	20
PubMed	(TILDA OR "Irish Longitudinal Study on Ageing") AND ("chronic disease" OR cardiovascular OR diabetes) AND (nutrition OR diet OR lifestyle OR "physical activity")	27
	(TILDA OR "Irish Longitudinal Study on Ageing") AND ("machine learning" OR "predictive model" OR "risk prediction")	9

Some search strings returned few or irrelevant results, particularly for ML related TILDA studies, highlighting a key research gap. Iterations continued until suitable combinations consistently returned relevant studies. Duplicates were removed using Zotero's reference management feature. The remaining studies were individually and manually checked for relevance to "ageing," "chronic disease," and ML. Using Zotero's notes functionality, the initially created table captured key elements for each paper, including model type, features used, and main findings for each paper in the collection, as illustrated in the PRISMA flow diagram (Figure 2.2).

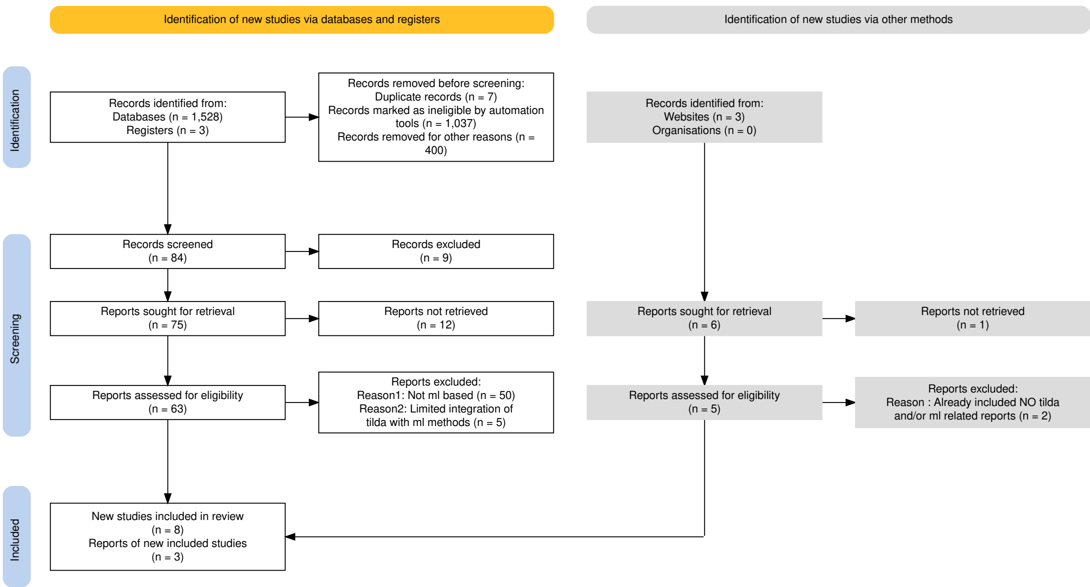

Figure 2.2: PRISMA chart for SRs showing the study selection process for the systematic mapping and integrative review [3].

2.1.2 Quality and relevance evaluation

To systematically assess the quality of the included studies, a customisable version of Newcastle-Ottawa Scale (NOS) was applied, Table 2.2. The NOS evaluates studies across three domains: Selection (S1–S4), Comparability (C1–C2), and Outcome (O1–O3). Selection criteria assess the representativeness of the sample, clarity of exposure measurement, and temporal alignment of outcomes. Comparability examines the control of key confounders, while Outcome evaluates measurement validity, follow up adequacy, and completeness. Each domain is scored 0–1, recall Bernoulli random variable (0 = criterion not met, 1 = criterion met), and the total score reflects the overall study quality.

This initial table guided a further refinement of the results into two sub tables: (1) studies critically related to TILDA, (2) studies using TILDA data with ML for predictive contexts, which revealed minimal to no integration with TILDA.

Table 2.2: Quality evaluation of Systematic Literature Review (SLR)s [2]

Reference	**S1**	**S2**	**S3**	**S4**	**C1**	**C2**	**O1**	**O2**	**O3**	**Total**
(Xu, 2023 [20])	1	1	1	1	1	1	1	1	1	9
(Donoghue, 2023 [15])	1	1	1	1	1	1	1	0	1	8
(Moguilner, 2021 [14])	1	1	1	1	1	0	1	1	1	8
(Davis, 2021 [4])	1	1	1	1	1	0	1	1	0	7
(Boyle, 2021 [21])	1	1	1	1	1	0	1	0	1	7
(Tzemah-Shahar, 2022 [22])	1	1	1	0	1	0	1	1	1	7
(A. Shirsath, 2024 [23])	1	1	1	1	1	1	1	0	1	8
(Romero-Ortuno, 2021 [24])	1	1	1	1	1	1	1	1	1	9
(Bardon, 2020 [25])	1	1	1	0	1	0	1	1	1	7
(Zhang, 2025 [26])	1	1	1	1	1	1	1	0	1	8
(Sutin, 2023 [27])	1	1	1	1	0	0	1	1	1	7

Table 2.3 summarises selected studies on ageing using TILDA and ML, reflecting key gaps and opportunities identified during systematic mapping. These steps were aligned with the research matrix created at the beginning, ensuring that the refinement of studies and synthesis of findings stayed closely tied to the original research goals.

2.1.3 Synthesis and Integration

Using the NOS scores, I annotated each paper in Zotero with dataset details, key variables, analytical approach, and main findings to ensure a consistent, critical appraisal. For the systematic component, I systematically extracted and synthesised data to identify trends (e.g., Markov models, logistic regression, 5-fold Cross Validation (CV)) and research gaps [2], complemented by DoctoralWriting resources [28, 29], which clarified how to define research questions and develop logical arguments.

The integration of a research matrix with systematic mapping and integrative synthesis ensured that the methodology was well suited to capturing the complexity of ageing, nutrition, chronic disease risk and predictive modelling studies with and without TILDA.

Table 2.3: Summary of selected studies on ageing using TILDA

Theme	Author/year	Dataset	Variables/features	Methods/model type	Key Findings/outcome	Limitations
FR	Romero-Ortuno (2021) [24]	Wave 1–5	FR state, FR phenotype (FP) components (Exhaustion(EX), Unexplained weight loss(WL), Weakness(WK), Slowness(SL), Low physical activity(LA))	Multi state Markov model, longitudinal 8 year follow up	32% of pre frail to non frail; FR strongly predicted 31% mortality	Self report bias; observational design
Falls + ML	Donoghue (2023) [15]	Wave 1–3	Gait, mobility, medications, history of falls	Conditional inference forest; 5-fold CV; undersampling for imbalance	Accuracy up to 69.7%; for adults aged 50–64.	Area Under the Receiver Operating Characteristic Curve (AUROC) values were <= 0.69 and PRAUC <= 0.39 indicating low predictive accuracy; no external validation; cross sectional; Class imbalance
Fear of Falling (FOF)	Zarudsky (2014) [30]	Wave 1	Age, sex, fall history	Cross sectional prevalence analysis	23.3% prevalence; higher in women; 70% with FOF had no recent falls in the last 12 months	Self reported; cross sectional
Functional Mobility	Huang (2019) [31]	Wave 1–2	Measured by Timed Up and Go (mobility test) (TUG), Hand Grip Strength (HGS), physical activity, vision, Body Mass Index (BMI), age	Linear regression, stratified analyses	Age modifies effects; PA and grip protect mobility; vision benefit only in middle aged (Beta = -0.032, P = 0.002)	Observational; 2 year follow up
Depression	Jacob (2023) [32]	Wave 1–2	Falls, FOF, HADS-A, Center for Epidemiologic Studies Depression Scale (CES-D)	Prospective logistic regression	Falls predicted incident anxiety Odds Ratio (OR)=1.58 and depression OR=1.43	Self reported falls; short follow up; mediation effect partially explored
	Laird (2023) [13]	Wave 1–5	Moderate-to-Vigorous Physical Activity (MVPA), walking, sociodemographics	Longitudinal regression; mixed model	Higher physical activity reduced depressive symptoms; dose response effect	Self reported PA; observational; limited causal inference
Nutrition	Murphy (2023) [9]	Wave 1–5	Plasma lutein, zeaxanthin	Linear/ordinal logistic regression	Higher carotenoids linked to slower FR progression Lower odds of becoming frail (ORs = 0.57–0.89)	Observational; not predictive of longitudinal muscle changes
	Bardon (2020) [25]	Wave 1–2	Hospitalisation, functional status, SOC support, sex, falls	Logistic regression; sex stratified analyses	Functional decline, hospitalisation, poor support increased malnutrition risk; sex differences	Observational(short follow up (2 years)); limited generalisability ¿65 years, self reported
ML	Xu (2023) [20]	Wave 1–3	105 predictors from 40,000+ variables (SOC, clinical, mental, family)	Stacked ensemble: RF, XRT, GLM, GBM, Deep Neural Network (DNN); Shapley Additive Explanations (SHAP) explainability	AUC=0.854 for 2 year Diabetes Mellitus (DM) prediction; top predictors LDL, cirrhosis, sex	No external validation; short follow up; ensemble complexity limits deployment
	Davis (2021) [4]	Wave 3	Age, HGS, chair stands, BMI, cognitive tests	Histogram gradient boosting regression; SHAP	r^2 test: UGS≈0.41, Maximum Gait Speed (MGS)≈0.46, GSR≈0.21	Modest predictive power; cross sectional; no external validation
	Zhang (2025) [26]	Longitudinal IMAGEN cohort	Psychopathology, personality, cognition, substance use, environment	Regularised logistic regression; diagnostic	Longitudinal AUC: ED=0.71, MDD=0.64, AUD=0.67; shared transdiagnostic predictors	Limited age range; no external replication
	Boyle (2021) [21]	Wave 3	Brain predicted age difference (brainPAD); processing speed; attention; cognitive flexibility; verbal fluency	Linear regression, correlation analyses	Higher brainPAD associated with lower cognitive performance across multiple domains	Cross sectional; no causal inference; limited sample

2.2 Frailty and physical function

FR reflects declining physiological reserve, providing a framework for understanding physical vulnerability in ageing. In TILDA, it is captured through both the FR phenotype and the FR index, with UGS, HGS, and mobility decline serving as practical predictors. These measures strongly predict hospitalisation and mortality.

2.2.1 Probabilistic interpretation of frailty

One can argue that from a probabilistic perspective, the chance of an adverse outcome (such as mortality, Y) given observed features (such as FR markers, X) as a conditional probability $P(Y \mid X)$. This allows us to deduce which FR features are most predictive of adverse outcomes. Features may be independent, dependent, or mutually exclusive.

The presence or absence of metabolic syndrome represents a mutually exclusive event that can inform risk models.

$$P(X_i \cap X_j) = P(X_i)P(X_j)$$

for independent features X_i and X_j, guiding feature selection and reducing redundancy in model inputs. This reflects the likelihood that an individual with certain FR characteristics will experience a particular outcome.

Evidence suggests that FR is dynamic, not irreversible. For example, multi state modelling showed that many prefrail adults transition back to non frail, although established FR strongly predicts mortality [24]. Metabolic syndrome, on the other hand, speeds up the onset of FR, which suggests a metabolic pathway of vulnerability. [33]. The presence of metabolic syndrome modifies the conditional probability $P(\text{FR} \mid \text{Metabolic Syndrome})$ and may increase the likelihood of adverse outcomes over time. Another detail is that age matters: obesity predicts mobility decline earlier in life, whereas later in life, traits that protect against this, including HGS, become more significant [31].

These findings partly conflict: FR appears both reversible and progressive, depending on risk context. This complexity highlights the need for probabilistic models that can update beliefs about risk as new features are observed, as formalised by Bayes theorem:

$$P(Y \mid X) = \frac{P(X \mid Y)P(Y)}{P(X)}$$

where $P(Y \mid X)$ is the posterior probability of outcome Y given features X.

2.2.2 Bayesian updating of frailty risk

Causal inference is limited by observational designs, and predictors are frequently examined separately, ignoring interactions. Most FR models have only been with the TILDA dataset and with this present study focuses on the same TILDA but with an expansion on the harmonised set to explore more longitudinal prediction within this context. These gaps point to the need for multidimensional approaches. Traditional regression models may not fully capture age dependent interactions, suggesting a role for more flexible methods such as machine learning techniques and models.

2.3 Mental health in older adults

MH, including depression, anxiety, loneliness, and SOC isolation, strongly shapes quality of life. Evidence from TILDA shows strong links to both physical and SOC factors. [34] found that participation in physical, SOC, and cognitive activities summarised by the Act-Belong-Commit (ABC) indicator was associated with lower depression and anxiety. This suggests SOC integration and lifestyle protect mental wellbeing. However, [35] reported that self reported measures may exaggerate such associations..., raising concerns of bias. These two findings conflict: the effect of SOC engagement is strong when self reported, weaker when objectively assessed.

Objective data strengthen the case for activity. Accelerometer based measures show a dose response link between moderate to vigorous activity and lower depressive symptoms [10]. Yet reverse pathways are also evident: incident falls and functional decline predict later depression, refereed by FOF [32]. Together, these results reveal a bidirectional relationship, MH both influences and is influenced by physical decline. Most studies rely on broad self report scales such as

CES-D or Hospital Anxiety and Depression Scale (HADS), with little biomarker or clinical validation.

This literature suggests MH in ageing cannot be modelled as a one way effect. It requires longitudinal, multidimensional approaches that can capture mutual pathways, an area where machine learning is well suited especially when combined with probabilistic models that estimate the likelihood and posterior probability of MH outcomes given multiple interacting features.

For example, we can estimate:

$$P(\text{Depression} \mid \text{Activity, Falls, SOC Engagement}),$$

quantifying how these interacting features together influence MH outcomes. Probabilistic modelling helps determine which features have the strongest effect and which may be redundant for predictive purposes.

2.4 Nutrition and biological ageing

Nutrition interacts with both biological and psychosocial systems, shaping ageing trajectories and FR. Biomarkers like vitamin D, lutein, and zeaxanthin have been linked to physical and cognitive outcomes in TILDA. [9] showed that higher plasma lutein and zeaxanthin dietary carotenoids found in leafy vegetables were linked to slower FR progression. By contrast, [36] found vitamin D deficiency predicted DM and musculoskeletal decline, especially among adults with low sunlight exposure. These studies point to different nutritional mechanisms: antioxidant protection versus metabolic regulation.

2.4.1 Bayesian perspective on nutritional biomarkers

The effect of nutrition on ageing outcomes can be formalised probabilistically using Bayes theorem:

$$P(\text{Outcome} \mid \text{Nutritional Status}) = \frac{P(\text{Nutritional Status} \mid \text{Outcome})P(\text{Outcome})}{P(\text{Nutritional Status})},$$

where the likelihood function $P(\text{Nutritional Status} \mid \text{Outcome})$ quantifies how observed nutritional biomarkers relate to health states, and the posterior $P(\text{Outcome} \mid$

Nutritional Status) captures updated beliefs about outcomes, including uncertainty in small sample studies. This helps identify which biomarkers are most informative for predicting ageing outcomes which supports feature selection in ML models.

2.4.2 Contextual and integrative considerations

Malnutrition also reflects SOC and functional risks. [25] found that hospitalisation, functional loss, and poor SOC support increased malnutrition risk, with notable sex differences. This highlights a limitation in much nutritional research: biological markers are studied without integrating SOC context.

The difference between biological and chronological age is a major point of debate. Biomarker based metrics that better capture these cumulative exposures, like FR indices or epigenetic clocks, frequently vary from chronological age. McCarthy and Azizi link metabolic syndrome and nutritional deficits to accelerated biological ageing [37, 38]. Although the majority of the evidence is cross sectional and has little ability to capture changes over time, these indicators offer a more precise prediction of vulnerability. Reliability is also decreased in biomarker research with small sample sizes, which can be formalised through the likelihood function $P(\text{Data} \mid \theta)$ and its impact on posterior uncertainty in Bayesian inference.

Taken together, nutrition interacts with biological, metabolic, and SOC systems. Isolating single nutrients risks oversimplification. Integrative modelling that combines biomarkers, clinical factors, and SOC determinants supported by machine learning pipelines offers a more promising route. Models that estimate the joint distribution of features and outcomes can better capture the complex dependencies relevant to biological ageing.

2.5 Machine Learning in ageing and chronic disease

ML is increasingly applied in ageing research because it can capture non linear and multidimensional relationships that traditional regression models, such as logistic regression,often miss [20, 21].

2.5.1 Feature sets in ML studies

Studies using ML in ageing research include many types of predictors: clinical measures, sociodemographic factors, lifestyle habits, cognitive tests, and psychosocial variables. A recurring pattern, however, is that models often underline easily measured physical health data, such as comorbidity counts, mobility scores, or vital signs, while more complex but important predictors, including nutrition and SOC factors, are underrepresented [30]. This imbalance may limit the ability of models to capture the full complexity of ageing, especially considering that MH and nutrition exert independent and interacting effects on later life outcomes see (here). Models that focus only on readily available clinical data risk reproducing the same biases seen in traditional regression approaches.

2.5.2 ML models used

The methodological approaches used in ageing research span both traditional and advanced models. Logistic regression and Markov models remain common baselines due to their interpretability and simplicity, yet they are often unable to capture non linearities or higher order interactions. More recent work has shifted towards ensemble methods such as RF and gradient boosting machines GBM, as well as DNNs, which generally achieve higher predictive accuracy.

Similarly, [39] evaluated mortality prediction using a standard logistic regression model with demographic, clinical, and functional predictors, achieving an AUC of 0.78, whereas, ensemble methods in [20] reached 0.854 for DM prediction, showing how flexible, non linear models can better capture complex interactions among ageing related predictors.

2.5.3 Performance and Validation

Predictive performance across studies is generally moderate to strong, with AUC values between 0.70 and 0.90 for outcomes like DM, UGS, depression, and mortality. This indicates that models can distinguish reasonably well between individuals who will or will not experience an outcome.

However, there are important limitations. Many studies rely only on internal validation, testing models on the same data used for training. This can inflate performance estimates and reduce confidence in applying the model to new populations. To add, Most studies are cross sectional, so causal or temporal effects cannot be assessed. When combined with the underuse of nutrition and SOC predictors, these issues indicate that current models may not fully represent the multidimensional nature of ageing and could produce context specific results rather than generalisable predictions [30].

2.5.4 Explainability in ML for ageing

As ML models become more complex, interpretability emerges as a critical challenge. Explainability tools like SHAP can highlight feature importance, but their use is inconsistent, particularly in deep learning models [4, 20, 40]. These methods improve transparency and can highlight unexpected drivers of outcomes, such as psychosocial variables being stronger predictors than biomedical ones, yet black box concerns remain, especially for deep learning models, where feature interactions are difficult to track. For ageing research, where clinical adoption requires both accuracy and trust, explainability is not optional but essential. Current efforts show promise but remain fragmented, with no consensus on best practice. This gap links directly to the broader methodological limits discussed in the next section on gaps in the literature.

2.6 Gaps in the Literature

First, most studies rely on TILDA or similar cohorts of Irish and predominantly older adults (50+). This limits confidence in how well results transfer to other populations. The present project focuses on TILDA to study ageing in depth but also makes use of the harmonised dataset, which links TILDA with comparable UK and US cohorts. This helps place findings in a broader international context while retaining a core emphasis on older adults.

Second, there is an over reliance on physical health predictors such as comorbidity counts and mobility measures, while nutrition, SOC, religious or spiritual, and environmental variables are often overlooked. As shown in [9], nutritional biomarkers like lutein and zeaxanthin are strong predictors of FR, yet these are rarely integrated into broader ML pipelines. By contrast, studies such as [25] highlight the role of SOC support and functional decline in malnutrition risk. Together, these examples show that excluding nutrition and SOC factors risks producing models that reinforce the same biomedical biases seen in regression heavy work. From a Bayesian viewpoint, the omission of key features reduces the explanatory power of the model and weakens the posterior probability $P(\text{Outcome} \mid \text{All Features})$ compared to $P(\text{Outcome} \mid \text{Limited Features})$. Including a richer set of features increases the likelihood that the model accurately reflects the true data generating process.

Third, methodological choices remain conservative. Regression models still dominate, while more advanced approaches such as DNNs or transformers are rarely explored. For instance, [39] achieved an AUC of 0.78 using logistic regression for mortality prediction in TILDA, whereas [20] reported an AUC of 0.854 with ensemble methods for DM incidence. This contrast demonstrates both the dominance of regression and its weaker performance compared to flexible ML models. Where ensemble or brainPAD approaches have been attempted, validation is almost always internal, with little cross cohort testing. Weak validation and limited feature sets reduce the generalisability of the estimated conditional probabilities and increase the risk of overfitting. Maximum likelihood estimation and Bayesian inference can provide a formal framework for assessing model fit and updating predictions as more diverse and representative data become available.

Fourth, many outcomes remain underexplored. Studies on dementia or FR frequently focus on cognitive or physical decline while leaving aside outcomes critical to older adults, such as pain, sleep quality, and subjective quality of life. For example, [9] used cross sectional plasma biomarkers to study FR but did not assess sleep or pain, while UGS models such as [4] similarly neglect subjective quality of life. This shows how reliance on narrow outcomes leaves broader aspects of wellbeing unmeasured.

Fifth, cross sectional bias and data handling weaken reliability. Much nutritional work, such as [9], relies on single time point biomarker data, which cannot capture longitudinal change. By contrast, studies applying multi wave TILDA data [24] demonstrate the added value of modelling temporal dynamics. Similarly, missing data are often handled through list wise deletion, as in [39], which risks biasing results toward healthier participants, whereas more robust imputation approaches remain underused. Missing or poorly handled data can distort the likelihood function and hence the posterior predictions, reducing confidence in the model's outputs.

Finally, interpretability challenges persist. While tools like SHAP have been applied to highlight feature importance [20, 40], their use is inconsistent, and consensus on best practice has yet to emerge. Without transparent explanations, even accurate models may face barriers to clinical acceptance. Taken together, these examples show that ageing research continues to rely on narrow predictors, conservative models, and weak validation. Addressing these gaps requires integrative, diverse, and transparent approaches that link multidimensional predictors with explainable ML.

2.7 Chapter Summary

This chapter reviewed evidence on ageing and chronic disease across three domains, FR and physical function, MH, and nutrition and biological ageing. We also examined how machine learning has been applied to predict later life outcomes. The findings showed that while FR is partly reversible, it remains strongly tied to mortality; that MH both shapes and is shaped by physical decline; and that nutrition interacts with biological, metabolic, and SOC systems in ways that are often underestimated.

While machine learning has improved prediction accuracy in ageing research, challenges remain in integrating diverse predictors and ensuring interpretability and longitudinal validation.

Addressing these challenges requires the integrative, diverse, and transparent approaches described in this thesis. These gaps motivate the methodological choices and implementation details outlined in Chapter 4.

Chapter 3

Technical Approach

3.1 Study Design

TILDA provides longitudinal data across five waves for community dwelling adults in Ireland. This thesis uses data from five core TILDA waves (Waves 1–5). For contextual comparison and benchmarking against international cohorts (see Chapter 1) however, it is important to note this is an external resource and not the output of the thesis's primary preprocessing pipeline. The primary analytical work focuses on the full five wave TILDA dataset.

3.2 Cohort Definition

All participants who provided data in at least one TILDA wave were considered eligible. To maximise the available sample for machine learning, no exclusions were applied on the basis of age, medical conditions, mobility, cognitive status, or health centre attendance.

Participants were removed only when, after the combined manual filtering and automated preprocessing pipeline, they had no remaining usable variables. Filtering therefore occurred primarily at the *variable level*, with features removed for extreme missingness, lack of variation, or non informative structure (see Chapter 4). This approach preserves the largest possible sample while ensuring all retained inputs are analytically meaningful.

3.2.1 Cohort inclusion criteria

- **Participant inclusion:** All respondents who contributed data in at least one wave were included in the initial sample.
- **Usable data requirement:** Participants were excluded only when, after manual filtering and preprocessing, no usable variables remained.
- **Longitudinal retention:** Individuals were retained regardless of how many waves they contributed to, enabling initial maximum dataset utilisation.
- **No clinical exclusions:** No participant was removed due to medical diagnoses, cognitive scores, functional limitations, or non attendance at the health centre assessment.

This approach prioritises data integrity and sample size for machine learning applications rather than clinical eligibility criteria common in classical epidemiological analyses. Figure 3.1.

3.3 Data Collection

Data used in this thesis were collected across five waves:

- **Wave 1:** 2009–2011
- **Wave 2:** 2012–2013
- **Wave 3:** 2014–2015
- **Wave 4:** 2016–2018
- **Wave 5:** 2018–2020

Each wave included three primary components:

1. **Computer Assisted Personal Interview (CAPI):** Home based, computer assisted personal interviews

2. **Self Completion Questionnaire (SCQ):** A paper based self completion survey focused on mental health, lifestyle, and subjective well being

3. **Health Assessment:** Clinical and biomarker measurements in centers or at home

3.3.1 Data Overview

Before any processing, all raw TILDA datasets were inspected manually and filtering was conducted in Excel to remove clearly non informative variables (administrative identifiers, constant columns, and variables with extreme missingness). Table 3.1 reports the raw dimensions (rows = participants; columns = variables) prior to filtering.

Table 3.1: Dimensions of Raw TILDA and Harmonised Datasets (Pre Filtering)

Dataset	Participants (N)	Raw Variables (N)
Wave 1	8501	1619
Wave 2	7206	724
Wave 3	6397	1029
Wave 4	5713	990
Wave 5	4978	983
Harmonised RAND	8501	572

3.3.2 Variable filtering criteria

The manual checks applied were rules-of-thumb used to rapidly triage variables before automated cleaning:

- **Missingness:** Drop variables with $>$ 80% missing values.

- **Unique values:** Drop variables with a single unique value; flag small cardinality variables (2–10 unique values) for retention.

- **Dominant category:** Flag variables where the top category frequency is extremely high (e.g., $>$ 90%) for manual review.

- **Datatypes / administrative fields:** Drop obvious identifiers and administrative labels (IDs, version tags, weights) unless required for linking.

These manual steps produced a reduced variable set, which was subsequently processed by the automated Six Step Cleaning Engine (see Chapter 4).

3.3.3 Variable filtering flowchart

Figure 3.1 shows the schematic of the manual filtering decisions and the intended placeholders for variable counts remaining after each step.

3.4 Thematic Grouping

After the manual inspection described above, the variable counts per wave were reduced. Table 4.1 reports the variable counts after manual filtering and prior to the automated cleaning pipeline.

To reduce dimensionality and improve interpretability, features were organised into six thematic domains consistent with ageing literature and the study aims. Table 3.2 summarises these domains and representative variables.

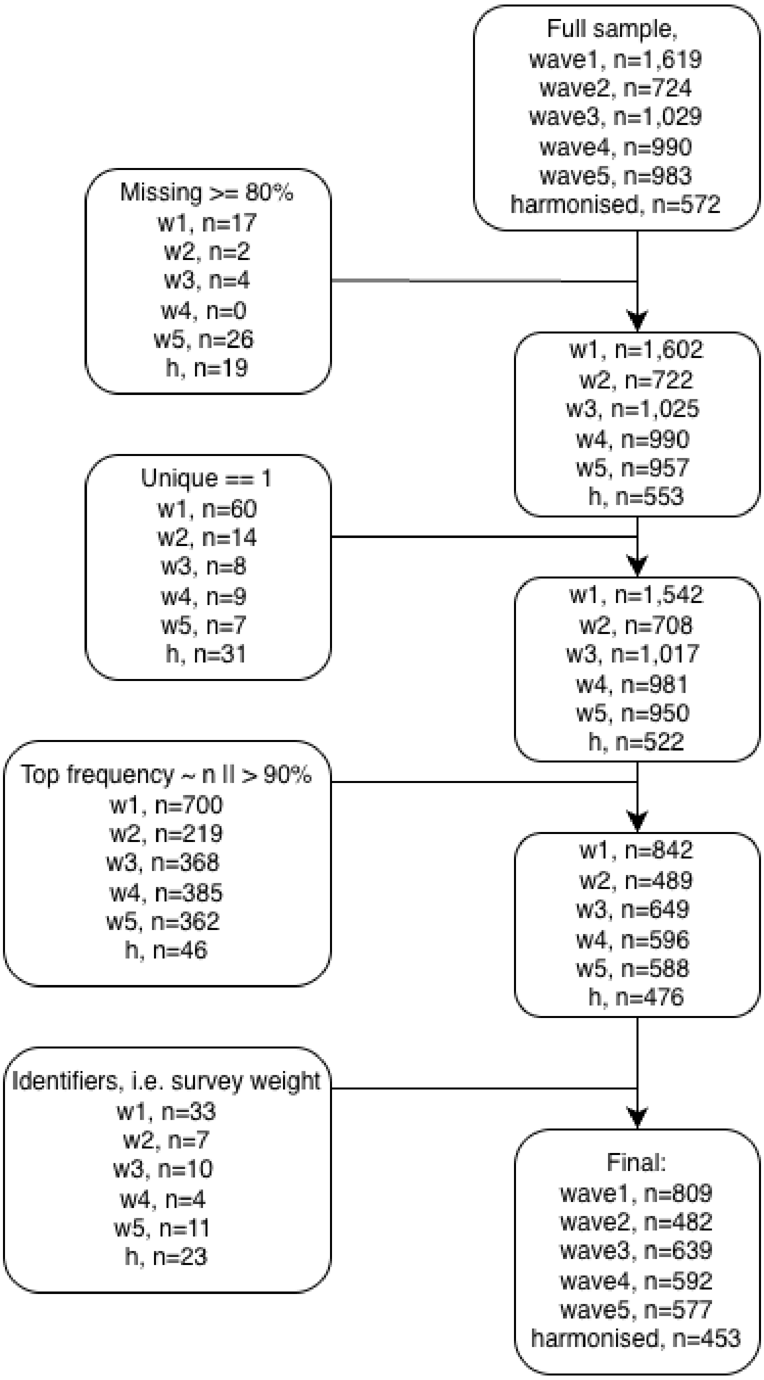

Figure 3.1: Flowchart of the manual variable filtering criteria. Placeholder counts indicate the number of variables remaining after each filtering step for each dataset.

Table 3.2: Thematic grouping of longitudinal features

ID	**Thematic Group**	**Description**	**Representative Variables**
1	Socio Demographic & Context	Baseline stratification factors	Age, Gender, Education, Income Variables (INC), Marital Status SOC
2	Physical Health & Clinical Baseline	Self reported and clinical indicators	Self Rated Health Screening Tests (SCR), BMI, BP, Comorbidity Count Medication Data (MD)
3	Functional Status & FR	Measures of mobility and physical decline	FR Index FR, UGS, Falls History *SCQFalls*, Instrumental Activities of Daily Living (IADL) Disability, Functional Impairment and Helpers (DIS)
4	Lifestyle & Behavioural	Modifiable health related behaviours	Smoking, Alcohol Intake *SCQAlco*, PA International Physical Activity Questionnaire (IPAQ)
5	Nutrition & Metabolic Biomarkers	Objective nutritional and metabolic markers	Plasma Carotenoids, Lutein, Zeaxanthin, Vitamin D/B12, Cholesterol
6	Psychological & Cognitive	Mental health and cognitive performance	CES-D, Loneliness *MHucla*, Scales and Questions on Cognition (COG), Choice Reaction Time (Cognitive Test) (CRT)

Chapter 4

Implementation Details

4.1 Data Preprocessing

The preprocessing pipeline primary aim was to convert all features, regardless of their original datatype (float, categorical, object, or mixed formats), into a standardised numerical representation suitable for machine learning. All non response and sentinel values were consistently mapped to the numerical placeholder `-1.0`, and all valid entries were converted to `float64`. This ensured compatibility across the TILDA waves and enabled uniform downstream operations for feature engineering and modelling.

4.1.1 Input data dimensions (post manual filtering)

The dimensions of the triaged datasets, which serve as the input for the automated cleaning engine, are reported below. This initial manual filtering step removed non informative variables (e.g., columns with zero variation, unique participant IDs, non essential administrative fields).

4.1.2 Cleaning Engine

The core challenge in converting heterogeneous variable types across five waves into a single numerical standard required constructing a reproducible pipeline to achieve *numerical homogeneity*. This was accomplished by developing a two

Table 4.1: Input Dimensions for Cleaning Pipeline (Post Manual Variable Filtering)

Dataset	Row (participants)	Column (variables)
Wave 1	8501	809
Wave 2	7206	482
Wave 3	6397	639
Wave 4	5713	592
Wave 5	4978	577
Harmonised RAND	8501	453

stage system: **Variable scenario mapping** and the **Six step deterministic cleaning engine**.

Stage 1: Variable scenario mapping and validation

An exploratory visualiser was used to systematically inspect the raw data distributions, unique values, and datatype inconsistencies across all five waves. This audit initially identified between five and seven distinct data scenarios, which were expanded and refined during development to nine total scenarios to cover the full scope of data heterogeneity: from simple numeric fields and numeric data with symbols, to likert type scales, calendar based frequency strings, and complex mixed string numeric fields. These nine initial scenarios were then systematically collapsed and categorised into the six final cleaning steps, reflecting an ordering based on transformation confidence.

The pipeline's first stage utilised an automated function, powered by the `re` (Regex) python library, to classify every raw variable into one of these six final scenarios. During development, a dedicated "unknown complex" category was crucial: variables routed here indicated a failure of the detection logic. This iterative debugging process, which identified and fixed classification errors, significantly drove the robustness of the final detection mechanism by ensuring the pipeline correctly classified variables that initially appeared ambiguous.

To ensure robustness, the classification output was immediately processed by a validation routine that generated a comprehensive dictionary of all variables, specific to its respective grouped detected scenario after each step (used for manual inspection), their detected scenario, and a sample of transformed values. This dictionary was formatted into a ".md" file for rapid, surface level verification, which honed the final mapping dictionary. This process of using automated detection followed by visual, manual validation was essential for correcting errors in classification logic before the subsequent scenario transformations were executed in the pipeline.

Stage 2: Deterministic six step cleaning sequence

The nine input scenarios were processed by the *Six step cleaning engine*. Resolving simple and unambiguous cases first increased the proportion of clean numerical variables before addressing complex structures, which involved the censored and multi-class scenarios (Figure 4.1). This deterministic sequence applied the same logic across all waves to ensure consistent data alignment.

The steps, and their technical focus, are:

1. **Empty / Blank Values:** Converts all `NaN`, empty strings, and whitespace to the standard non response value `-1.0`.

2. **Sentinel Values:** Standardises wave specific non response codes (e.g., `-98`, `-99`) and documented textual sentinels (e.g., `"dk"`, `"not applicable"`) from the TILDA Pseudonymised Microdata File (PMF) release guide to the unified `-1.0` placeholder.

3. **Numeric Transformation:** Removes non numeric characters (units, symbols, white spaces) and converts censored or annotated numeric strings (e.g., `"50+"`, `"<10"`, `"1-2.5"`) to a numerical float equivalent (e.g., boundary value or midpoint).

4. **Censored Categorical:** Applies explicit, known ordinal transformations and pre defined mappings to variables where text labels represent a latent numerical or ordered scale (e.g., health rating scales, likert ranges). This step was intentionally performed manually for a subset of low count variables to build intuition and reliable mapping dictionaries for the more

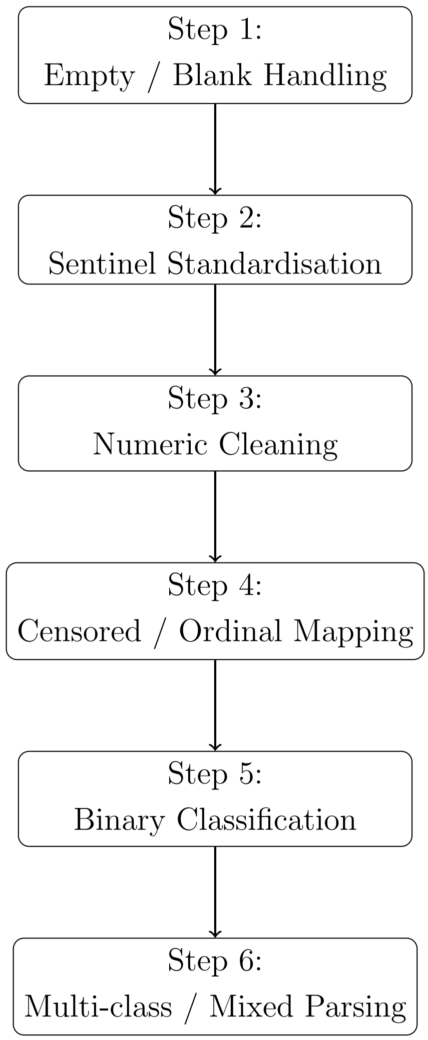

Figure 4.1: Flowchart of the Six Step Cleaning Pipeline applied to all TILDA waves. See full transformation tables in Appendix A.7 for detailed scenario progression at each step.

complex subsequent scenarios. This included conversions for calendar based frequency strings like `"twice a week"` to a numerical weekly equivalent.

5. **Binary Categorical:** Automatically maps two level variables (e.g., Yes/No, Male/Female) to `1.0/0.0`. The mapping relies on *semantic detection rules* (e.g., assigning `0.0` if the string contains `"not"` or `"no"` (context constrained) or a wave specific refusal code) and *count detection rules* (where a variable has exactly two non missing unique values).

6. **Multi-class / Mixed (The Complex Scenario):** This final step processes all remaining multi label, mixed type, and string numeric hybrid columns. The challenge was more complex by the fact that the order of categorical string values for the same variable could differ across waves in some cases. To handle this, numerical assignments were made using an explicit dictionary lookup based on the original string, rather than relying on positional or alphabetical order. This ensured consistent numerical representation. The step performs extensive lexical processing, dedicated number extraction, and rule based mapping to convert dense, unstandardised text into numerical floats, ensuring all residual variable types are successfully quantified. The transformed values were subject to increased manual verification due to the variable complexity.

4.2 Data integration and longitudinal alignment

Following cleaning, features were aligned across waves based on:

Chapter 5

Empirical Evaluation

5.1 References

Open the file 'references_myphd' with BibTex. The file is located in '/7_references/'

The file contains a series of templates for the proper formatting of references according to the Harvard referencing style. Click on each reference in the text below to see how Latex has formatted the reference on the References section.

- **Book**: (? , p. 10) (page number required only if referring to an in-text quote)
- **Book Chapter**: (?)
- **Book Contribution**: (?)
- **MUSIC CD/DVD**: (?)
- **Journal Article**: (?)
- **Patent**: (?)
- **PhD/Master Dissertation**: (?)
- **Conference Paper**: (?)
- **Technical Report**: (?)

- **Unpublished Report**: (**?**)
- **Web Videos**(**?**) (. . . not the name of the person who uploaded the video though)
- **Web Paper**: (**?**)
- **Web Journal Article**: (**?**)
- **Webpage**: (**?**)
- **Downloaded Images or Sounds**: (**?**)

You can also you this kind of referencing if needed in the text:
Surname [**?**] said that . . .

Chapter 6

Conclusion

6.1 Summary

6.2 Conclusions

6.2.1 Data availability Statement

The data analysed in this study is subject to the following licenses/ restrictions: The datasets generated during and/or analysed during the current study are not publicly available due to data protection regulations but are accessible at TILDA on reasonable request. Requests to access these datasets should be directed to https://tilda.tcd.ie/data/accessing-data/.

6.2.2 Ethics Statement

The studies involving human participants were reviewed and approved by Faculty of Health Sciences Research Ethics Committee at Trinity College Dublin, Ireland. The patients/ participants provided their written informed consent to participate in this study.

6.3 Contributions

6.4 Future Work

Appendix A

Scenario Cleaning Pipeline Tables

Table A.1: Initial Raw Feature Counts Categorised by Nine Scenarios (Pre Cleaning)

Scenario	Wave 1	Wave 2	Wave 3	Wave 4	Wave 5	Harmonised
Empty / blank values	9	0	7	0	9	0
Numeric with sentinel	187	59	143	60	100	141
Clean numeric	14	27	18	22	20	8
Censored / bracketed numeric	84	71	91	85	85	73
Binary categorical	248	159	177	220	207	187
Multi class categorical / mixed	267	166	203	205	156	44
Single value categorical	0	0	0	0	0	0
Special symbol / mixed	0	0	0	0	0	0
Unknown / complex mixed	0	0	0	0	0	0
Total Columns	**809**	**482**	**639**	**592**	**577**	**453**

Table A.2: Scenario Count Progression: Post Step 1 (Empty / Blank Handling)

Scenario	Wave 1	Wave 2	Wave 3	Wave 4	Wave 5	Harmonised
Empty / blank values	0	0	0	0	0	0
Numeric with sentinel	81	31	42	32	75	15
Clean numeric	124	55	123	50	54	134
Censored / bracketed numeric	84	71	91	85	85	73
Binary categorical	251	159	179	220	207	187
Multi class categorical / mixed	269	166	204	205	156	44
Single value categorical	0	0	0	0	0	0
Special symbol / mixed	0	0	0	0	0	0
Unknown / complex mixed	0	0	0	0	0	0
Total	**809**	**482**	**639**	**592**	**577**	**453**

Table A.3: Scenario Count Progression: Post Step 2 (Sentinel Standardisation)

Scenario	**Wave 1**	**Wave 2**	**Wave 3**	**Wave 4**	**Wave 5**	**Harmonised**
Empty / blank values	0	0	0	0	0	0
Numeric with sentinel	0	0	0	0	0	0
Clean numeric	205	86	165	82	129	149
Censored / bracketed numeric	84	71	91	85	85	73
Binary categorical	251	159	179	220	207	187
Multi class categorical / mixed	269	166	204	205	156	44
Single value categorical	0	0	0	0	0	0
Special symbol / mixed	0	0	0	0	0	0
Unknown / complex mixed	0	0	0	0	0	0
Total	**809**	**482**	**639**	**592**	**577**	**453**

Table A.4: Scenario Count Progression: Post Step 3 (Numeric Cleaning)

Scenario	**Wave 1**	**Wave 2**	**Wave 3**	**Wave 4**	**Wave 5**	**Harmonised**
Empty / blank values	0	0	0	0	0	0
Numeric with sentinel	0	0	0	0	0	0
Clean numeric	263	117	215	124	172	149
Censored / bracketed numeric	16	23	17	16	17	73
Binary categorical	251	160	180	221	207	187
Multi class categorical / mixed	279	182	227	231	181	44
Single value categorical	0	0	0	0	0	0
Special symbol / mixed	0	0	0	0	0	0
Unknown / complex mixed	0	0	0	0	0	0
Total	**809**	**482**	**639**	**592**	**577**	**453**

Table A.5: Scenario Count Progression: Post Step 4 (Censored / Ordinal Mapping)

Scenario	**Wave 1**	**Wave 2**	**Wave 3**	**Wave 4**	**Wave 5**	**Harmonised**
Empty / blank values	0	0	0	0	0	0
Numeric with sentinel	0	0	0	0	0	0
Clean numeric	279	140	233	140	189	222
Censored / bracketed numeric	0	0	0	0	0	0
Binary categorical	251	160	180	221	207	187
Multi class categorical / mixed	279	182	226	231	181	44
Single value categorical	0	0	0	0	0	0
Special symbol / mixed	0	0	0	0	0	0
Unknown / complex mixed	0	0	0	0	0	0
Total	**809**	**482**	**639**	**592**	**577**	**453**

Table A.6: Scenario Count Progression: Post Step 5 (Binary Classification)

Scenario	**Wave 1**	**Wave 2**	**Wave 3**	**Wave 4**	**Wave 5**	**Harmonised**
Empty / blank values	0	0	0	0	0	0
Numeric with sentinel	0	0	0	0	0	0
Clean numeric	530	300	413	361	396	409
Censored / bracketed numeric	0	0	0	0	0	0
Binary categorical	0	0	0	0	0	0
Multi class categorical / mixed	279	182	226	231	181	44
Single value categorical	0	0	0	0	0	0
Special symbol / mixed	0	0	0	0	0	0
Unknown / complex mixed	0	0	0	0	0	0
Total	**809**	**482**	**639**	**592**	**577**	**453**

Table A.7: Final Dfs Cleaned: Post Step 6 (Multi-Class / Mixed Parsing)

Scenario	**Wave 1**	**Wave 2**	**Wave 3**	**Wave 4**	**Wave 5**	**Harmonised**
Empty / blank values	0	0	0	0	0	0
Numeric with sentinel	0	0	0	0	0	0
Clean numeric	809	482	639	592	577	453
Censored / bracketed numeric	0	0	0	0	0	0
Binary categorical	0	0	0	0	0	0
Multi class categorical / mixed	0	0	0	0	0	0
Single value categorical	0	0	0	0	0	0
Special symbol / mixed	0	0	0	0	0	0
Unknown / complex mixed	0	0	0	0	0	0
Total	**809**	**482**	**639**	**592**	**577**	**453**

Appendix B

Title

Type here you text as usual . . .

Appendix C

Code for estimating attitude

AHRS_TRIAD.C

```
/*
 *
 *  Copyright (c) 2013, Giuseppe Torre
 *  All rights reserved.
 *
 *  Redistribution and use in source and binary forms, with or without
 *  modification, are permitted provided that the following conditions are met:
 *      * Redistributions of source code must retain the above copyright
 *        notice, this list of conditions and the following disclaimer.
 *      * Redistributions in binary form must reproduce the above copyright
 *        notice, this list of conditions and the following disclaimer in the
 *        documentation and/or other materials provided with the distribution.
 *      * Neither the name of the BREAKFAST LLC nor the
 *        names of its contributors may be used to endorse or promote products
 *        derived from this software without specific prior written permission.
 *
 *  THIS SOFTWARE IS PROVIDED BY THE COPYRIGHT HOLDERS AND CONTRIBUTORS "AS IS" AND
 *  ANY EXPRESS OR IMPLIED WARRANTIES, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED
 *  WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE ARE
 *  DISCLAIMED. IN NO EVENT SHALL BREAKFAST LLC BE LIABLE FOR ANY
 *  DIRECT, INDIRECT, INCIDENTAL, SPECIAL, EXEMPLARY, OR CONSEQUENTIAL DAMAGES
 *  (INCLUDING, BUT NOT LIMITED TO, PROCUREMENT OF SUBSTITUTE GOODS OR SERVICES;
 *  LOSS OF USE, DATA, OR PROFITS; OR BUSINESS INTERRUPTION) HOWEVER CAUSED AND
 *  ON ANY THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT
 *  (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE OF THIS
 *  SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.
 *
 */



#include "ext.h"
#include "ext_mess.h"
#include <string.h>
#include <stdio.h>
#include <math.h>
#define EPSILON 1e-6

//******************CALIBRATION STUFF*************************//
/* INTRODUCE THE OFFSET VALUES IN THE FOLLOWING VARIABLES IN ADC values */
#define ACCELX_OFFSET 2043 //initial offset in accelerometer X
#define ACCELY_OFFSET 2053 // initial offset in accelerometer Y
#define ACCELZ_OFFSET 2264//initial offset in accelerometer Z

#define MagX_OFFSET 1917 //initial offset in Magnetometer X
```

Appendix

```
#define MagY_OFFSET 1813 // initial offset in Magnetometer Y
#define MagZ_OFFSET 1667//initial offset in Magnetometer Z
/* WE SHOULD CALIBRATE THE FOLLOWING VALUES FOR EACH PARTICULAR IMU axis */
#define ACCELX_Resolution 536
#define ACCELY_Resolution 661
#define ACCELZ_Resolution 559
#define MagX_Resolution 186
#define MagY_Resolution 189
#define MagZ_Resolution 170
//****************** END CALIBRATION STUFF  *************************//


void *this_class; // Required. Global pointing to this class

typedef struct _triad // Data structure for this object
{
        t_object m_ob;    // Must always be the first field; used by Max

        Atom m_args[9];   // we want our inlet to be receiving a list of 10 elements

        long m_value;     //inlet

        void *m_R1;               //these are all the outlets for the 3 X 3 Matrix
        void *m_R2;               // R1 --> firts top left cell ..R2 middle cell of the first
            row ...and so on
        void *m_R3;                                  //
        void *m_R4;                                  // End Outlets
            ----------------------------------------

} t_triad;

                        void *triad_new(long value);


                        void triad_assist(t_triad *triad, void *b, long msg, long arg, char *
                            s);
                        void triad_free(t_triad *triad);



                        void triad_list(t_triad *x, Symbol *s, short argc, t_atom *argv);




                        void MatrixByMatrix(double *Result,double *MatrixLeft,double *
                            MatrixRight);

                        void Matrix2Quat(double *Quat,double *Matrix);
                        void Quat2Matrix(double *Matrix, double *Quat);
                        void inverseQuat(double *InvQuat, double *RegQuat);
                        void NormQuat(double *YesQuat, double *NotQuat);
                        void Slerp(double *NewQuat, double *OldQuat, double *CurrentQuat);
                        void NormVect(double *YesVect, double *NotVect);
                        double orientationMatrix[9];
                        double Result[9];
                        int     i;
                        double InvorientationMatrix[9];

                        double temp[6];
                        double ref[6];
                        double vectAx[3];
                        double vectAy[3];
                        double vectAz[3];
                        double vectBx[3];
                        double vectBy[3];
                        double vectBz[3];
                        double MagnCrosProd_A;
                        double MagnCrosProd_B;
                        double accnorm, magnorm, earthnorm, VectAynorm, VectAznorm,
                            VectBynorm, VectBznorm;
```

```
                              double m[9], n[9];
                              double quat_e[4];
                              double invquat_e[4];
                              double mult;
                              double quat_new[4];
                              double quat_old[4];
                              double ecs,accex_ADCnumber,y,accey_ADCnumber,z,accez_ADCnumber,mx,
                                    magnx_ADCnumber,my,magny_ADCnumber,mz,magnz_ADCnumber;
// SLERP Variables
                              double trace, Suca;
                              double tol[4], omega, sinom, cosom, scale0, scale1, tez,
                                    orientationMatrixA[9];

int main(void)
{
          // set up our class: create a class definition
          setup((t_messlist**) &this_class, (method)triad_new, (method)triad_free, (short)
                sizeof(t_triad), 0L,A_GIMME, 0);
     addmess((method)triad_list, "list", A_GIMME, 0);
          addmess((method)triad_assist, "assist", A_CANT, 0);
          finder_addclass("Maths","triad");
          post("....I'm-TRIAD-Object!....from-AHRS-Library...",0);
          return 0;
}

/* ---------------triad_new ----------------*/

void *triad_new(long value)
{
          t_triad *triad;
          triad = (t_triad *)newobject(this_class);          // create the new instance and return
                 a pointer to it
          triad->m_R4 = floatout(triad);
          triad->m_R3 = floatout(triad);
          triad->m_R2 = floatout(triad);
          triad->m_R1 = floatout(triad);

return(triad);
}

etc.

etc.
```

References

[1] Kai Petersen, Sairam Vakkalanka, and Ludwik Kuzniarz. Guidelines for conducting systematic mapping studies in software engineering: An update. *Information and Software Technology*, 64:1–18, August 2015. ISSN 0950-5849. doi: 10.1016/j.infsof.2015.03.007. vi, 5, 7

[2] Barbara Kitchenham, O. Pearl Brereton, David Budgen, Mark Turner, John Bailey, and Stephen Linkman. Systematic literature reviews in software engineering – A systematic literature review. *Information and Software Technology*, 51(1):7–15, January 2009. ISSN 09505849. doi: 10.1016/j.infsof.2008.09.009. vi, 5, 9, 10

[3] Neal Haddaway, Matthew J Page, Chris C Pritchard, and Luke A McGuinness. PRISMA2020: An R package and Shiny app for producing PRISMA 2020-compliant flow diagrams, with interactivity for optimised digital transparency and Open Synthesis, 2022. vii, 8

[4] James R. C. Davis, Silvin P. Knight, Orna A. Donoghue, Belinda Hernández, Rossella Rizzo, Rose Anne Kenny, and Roman Romero-Ortuno. Comparison of Gait Speed Reserve, Usual Gait Speed, and Maximum Gait Speed of Adults Aged 50+ in Ireland Using Explainable Machine Learning. *Front Netw Physiol*, 1:754477, 2021. ISSN 2674-0109. doi: 10.3389/fnetp.2021.754477. 1, 9, 11, 17, 18

[5] Patrick V. Moore, Kathleen Bennett, and Charles Normand. Counting the time lived, the time left or illness? Age, proximity to death, morbidity and prescribing expenditures. *Social Science & Medicine*, 184:1–14, 2017. ISSN 0277-9536. doi: 10.1016/j.socscimed.2017.04.038.

[6] Ying Huang, Yuhao Su, Ying Jiang, and Meilan Zhu. Sex differences in the associations between blood pressure and anxiety and depression scores in a middle-aged and elderly population: The Irish Longitudinal Study on Ageing (TILDA). *Jour-*

nal of Affective Disorders, 274:118–125, September 2020. ISSN 0165-0327. doi: 10.1016/j.jad.2020.05.133. 1

[7] World Health Organization Europe. New data: Noncommunicable diseases cause 1.8 million avoidable deaths and cost US$ 514 billion every year, reveals new WHO/Europe report. https://www.who.int/europe/news/item/27-06-2025-new-data–noncommunicable-diseases-cause-1-8-million-avoidable-deaths-and-cost-us-514-billion-USD-every-year–reveals-new-who-europe-report, 2025. 1

[8] Niamh Rice and Charles Normand. The cost associated with disease-related malnutrition in Ireland. *Public Health Nutr*, 15(10):1966–1972, October 2012. ISSN 1475-2727. doi: 10.1017/S1368980011003624. 1

[9] Caoileann H. Murphy, Eoin Duggan, James Davis, Aisling M. O'Halloran, Silvin P. Knight, Rose Anne Kenny, Sinead N. McCarthy, and Roman Romero-Ortuno. Plasma lutein and zeaxanthin concentrations associated with musculoskeletal health and incident frailty in The Irish Longitudinal Study on Ageing (TILDA). *Experimental Gerontology*, 171:112013, 2023. ISSN 0531-5565. doi: 10.1016/j.exger.2022.112013. 1, 2, 11, 14, 18, 19

[10] Eamon Laird, Charlotte Lund Rasmussen, Rose Anne Kenny, and Matthew P. Herring. Physical Activity Dose and Depression in a Cohort of Older Adults in The Irish Longitudinal Study on Ageing. *JAMA Netw Open*, 6(7):e2322489, July 2023. ISSN 2574-3805. doi: 10.1001/jamanetworkopen.2023.22489. 1, 13

[11] C. P. McDowell, R. K. Dishman, M. Hallgren, C. MacDonncha, and M. P. Herring. Associations of physical activity and depression: Results from the Irish Longitudinal Study on Ageing. *Experimental Gerontology*, 112:68–75, 2018. ISSN 0531-5565. doi: 10.1016/j.exger.2018.09.004. 1

[12] Cathal McCrory, Ciaran Finucane, Celia O'Hare, John Frewen, Hugh Nolan, Richard Layte, Patricia M. Kearney, and Rose Anne Kenny. Social Disadvantage and Social Isolation Are Associated With a Higher Resting Heart Rate: Evidence From The Irish Longitudinal Study on Ageing. *J Gerontol B Psychol Sci Soc Sci*, 71(3):463–473, May 2016. ISSN 1079-5014. doi: 10.1093/geronb/gbu163. 1

[13] Eamon Laird, Matthew P. Herring, Brian P. Carson, Catherine B. Woods, Cathal Walsh, Rose Anne Kenny, and Charlotte Lund Rasmussen. Physical activity for

depression among the chronically Ill: Results from older diabetics in the Irish longitudinal study on ageing. *Psychiatry Research*, 326:115274, August 2023. ISSN 0165-1781. doi: 10.1016/j.psychres.2023.115274. 2, 11

[14] Sebastian Moguilner, Silvin P. Knight, James R. C. Davis, Aisling M. O'Halloran, Rose Anne Kenny, and Roman Romero-Ortuno. The Importance of Age in the Prediction of Mortality by a Frailty Index: A Machine Learning Approach in the Irish Longitudinal Study on Ageing. *Geriatrics (Basel)*, 6(3):84, August 2021. ISSN 2308-3417. doi: 10.3390/geriatrics6030084. 2, 9

[15] Orna A Donoghue, Belinda Hernandez, Matthew D L O'Connell, and Rose Anne Kenny. Using Conditional Inference Forests to Examine Predictive Ability for Future Falls and Syncope in Older Adults: Results from The Irish Longitudinal Study on Ageing. *J Gerontol A Biol Sci Med Sci*, 78(4):673–682, April 2023. ISSN 1758-535X. doi: 10.1093/gerona/glac156. 3, 9, 11

[16] Richard J. Torraco. Writing Integrative Literature Reviews: Guidelines and Examples. *Human Resource Development Review*, 4(3):356–367, September 2005. ISSN 1534-4843, 1552-6712. doi: 10.1177/1534484305278283. 5

[17] Tonette S. Rocco and Maria S. Plakhotnik. Literature Reviews, Conceptual Frameworks, and Theoretical Frameworks: Terms, Functions, and Distinctions. *Human Resource Development Review*, 8(1):120–130, March 2009. ISSN 1534-4843, 1552-6712. doi: 10.1177/1534484309332617. 5

[18] Andrew D Oxman and Gordon H Guyatt. Guidelines for reading literature reviews. 5

[19] Muhammad Azeem Ashraf, Meijia Yang, Yufeng Zhang, Mouna Denden, Ahmed Tlili, Jiayi Liu, Ronghuai Huang, and Daniel Burgos. A Systematic Review of Systematic Reviews on Blended Learning: Trends, Gaps and Future Directions. *Psychology Research and Behavior Management*, 14:1525–1541, October 2021. ISSN null. doi: 10.2147/PRBM.S331741. 5

[20] Xuezhong Xu, Xue Mingyang, Jie Yang, Hailong Zheng, and Zhifei Che. Tailored machine learning for evaluating the long-term diabetes risk in older individuals: Findings from the Irish Longitudinal Study on Ageing (TILDA). *BMJ Open*, 13 (5):e072991, May 2023. ISSN 2044-6055. doi: 10.1136/bmjopen-2023-072991. 9, 11, 16, 17, 18, 19

[21] Rory Boyle, Lee Jollans, Laura M. Rueda-Delgado, Rossella Rizzo, Görsev G. Yener, Jason P. McMorrow, Silvin P. Knight, Daniel Carey, Ian H. Robertson, Derya D. Emek-Savaş, Yaakov Stern, Rose Anne Kenny, and Robert Whelan. Brain-predicted age difference score is related to specific cognitive functions: A multi-site replication analysis. *Brain Imaging Behav*, 15(1):327–345, February 2021. ISSN 1931-7565. doi: 10.1007/s11682-020-00260-3. 9, 11, 16

[22] Roy Tzemah-Shahar, Hagit Hochner, Khalil Iktilat, and Maayan Agmon. What can we learn from physical capacity about biological age? A systematic review. *Ageing Research Reviews*, 77:101609, May 2022. ISSN 1568-1637. doi: 10.1016/j.arr.2022.101609. 9

[23] Morgana A. Shirsath, John D. O'Connor, Rory Boyle, Louise Newman, Silvin P. Knight, Belinda Hernandez, Robert Whelan, James F. Meaney, and Rose Anne Kenny. Slower speed of blood pressure recovery after standing is associated with accelerated brain ageing: Evidence from The Irish Longitudinal Study on Ageing (TILDA). *Cerebral Circulation - Cognition and Behavior*, 6:100212, January 2024. ISSN 2666-2450. doi: 10.1016/j.cccb.2024.100212. 9

[24] Roman Romero-Ortuno, Peter Hartley, James Davis, Silvin P. Knight, Rossella Rizzo, Belinda Hernández, Rose Anne Kenny, and Aisling M. O'Halloran. Transitions in frailty phenotype states and components over 8 years: Evidence from The Irish Longitudinal Study on Ageing. *Archives of Gerontology and Geriatrics*, 95: 104401, 2021. ISSN 0167-4943. doi: 10.1016/j.archger.2021.104401. 9, 11, 12, 19

[25] Laura A Bardon, Melanie Streicher, Clare A Corish, Michelle Clarke, Lauren C Power, Rose Anne Kenny, Deirdre M O'Connor, Eamon Laird, Eibhlis M O'Connor, Marjolein Visser, Dorothee Volkert, Eileen R Gibney, and MaNuEL Consortium. Predictors of Incident Malnutrition in Older Irish Adults from the Irish Longitudinal Study on Ageing Cohort—A MaNuEL study. *J Gerontol A Biol Sci Med Sci*, 75(2):249–256, January 2020. ISSN 1079-5006. doi: 10.1093/gerona/gly225. 9, 11, 15, 18

[26] Zuo Zhang, Lauren Robinson, Robert Whelan, Lee Jollans, Zijian Wang, Frauke Nees, Congying Chu, Marina Bobou, Dongping Du, Ilinca Cristea, Tobias Banaschewski, Gareth J. Barker, Arun L. W. Bokde, Antoine Grigis, Hugh Garavan, Andreas Heinz, Rüdiger Brühl, Jean-Luc Martinot, Marie-Laure Paillère Martinot, Eric Artiges, Dimitri Papadopoulos Orfanos, Luise Poustka, Sarah Hohmann,

Sabina Millenet, Juliane H. Fröhner, Michael N. Smolka, Nilakshi Vaidya, Henrik Walter, Jeanne Winterer, M. John Broulidakis, Betteke Maria van Noort, Argyris Stringaris, Jani Penttilä, Yvonne Grimmer, Corinna Insensee, Andreas Becker, Yuning Zhang, Sinead King, Julia Sinclair, Gunter Schumann, Ulrike Schmidt, and Sylvane Desrivières. Machine learning models for diagnosis and risk prediction in eating disorders, depression, and alcohol use disorder. *Journal of Affective Disorders*, 379:889–899, June 2025. ISSN 0165-0327. doi: 10.1016/j.jad.2024.12.053. 9, 11

[27] Dr Angelina R. Sutin, Martina Luchetti, Damaris Aschwanden, Yannick Stephan, Amanda A. Sesker, and Antonio Terracciano. Sense of meaning and purpose in life and risk of incident dementia: New data and meta-analysis. *Archives of Gerontology and Geriatrics*, 105:104847, 2023. ISSN 0167-4943. doi: 10.1016/j.archger.2022.104847. 9

[28] doctoralwriting. Creating the literature review: Research questions and arguments, February 2016. 10

[29] doctoralwriting. Creating the literature review: Research questions and arguments (Part 3 of 4), February 2016. 10

[30] A. A. Zarudsky and K. I. Prashchayeu. P277: One-legged balance test and falls in old patients with cardiovascular diseases. *European Geriatric Medicine*, 5:S171–S172, 2014. ISSN 1878-7649. doi: 10.1016/S1878-7649(14)70448-6. 11, 16, 17

[31] Chuanying Huang, Shuqin Sun, Xiaocao Tian, Tong Wang, Tianyu Wang, Haiping Duan, and Yili Wu. Age modify the associations of obesity, physical activity, vision and grip strength with functional mobility in Irish aged 50 and older. *Archives of Gerontology and Geriatrics*, 84:103895, 2019. ISSN 0167-4943. doi: 10.1016/j.archger.2019.05.020. 11, 12

[32] Louis Jacob, Karel Kostev, Jae Il Shin, Lee Smith, Hans Oh, Adel S. Abduljabbar, Josep Maria Haro, and Ai Koyanagi. Falls increase the risk for incident anxiety and depressive symptoms among adults aged ≥50 years: An analysis of the Irish longitudinal study on ageing. *Archives of Gerontology and Geriatrics*, 114:105098, 2023. ISSN 0167-4943. doi: 10.1016/j.archger.2023.105098. 11, 13

[33] Kevin McCarthy, Eamon Laird, Aisling M. O'Halloran, Padraic Fallon, Román Romero Ortuño, and Rose Anne Kenny. Association between metabolic

syndrome and risk of both prevalent and incident frailty in older adults: Findings from The Irish Longitudinal Study on Ageing (TILDA). *Experimental Gerontology*, 172:112056, 2023. ISSN 0531-5565. doi: 10.1016/j.exger.2022.112056. 12

[34] Ziggi Ivan Santini, Ai Koyanagi, Stefanos Tyrovolas, Josep Maria Haro, Robert J. Donovan, Line Nielsen, and Vibeke Koushede. The protective properties of Act-Belong-Commit indicators against incident depression, anxiety, and cognitive impairment among older Irish adults: Findings from a prospective community-based study. *Experimental Gerontology*, 91:79–87, 2017. ISSN 0531-5565. doi: 10.1016/j.exger.2017.02.074. 13

[35] A. L. Juola, M. P. Bjorkman, H. Kautiainen, S. Pylkkanen, U. H. Finne-Soveri, H. Soini, K. H. Pitkälä, and J. S. Bell. O1.17: Nursing staff education to reduce potentially harmful medication use among older people in assisted living facilities: Effects of randomized controlled trial on cognition and falls. *European Geriatric Medicine*, 5:S50–S51, 2014. ISSN 1878-7649. doi: 10.1016/S1878-7649(14)70100-7. 13

[36] Kevin McCarthy, Eamon Laird, Aisling M. O'Halloran, Cathal Walsh, Martin Healy, Annette L. Fitzpatrick, James B. Walsh, Belinda Hernández, Padraic Fallon, Anne M. Molloy, and Rose Anne Kenny. Association between vitamin D deficiency and the risk of prevalent type 2 diabetes and incident prediabetes: A prospective cohort study using data from The Irish Longitudinal Study on Ageing (TILDA). *EClinicalMedicine*, 53:101654, November 2022. ISSN 2589-5370. doi: 10.1016/j.eclinm.2022.101654. 14

[37] Kevin McCarthy, Aisling M. O'Halloran, Padraic Fallon, Rose Anne Kenny, and Cathal McCrory. Metabolic syndrome accelerates epigenetic ageing in older adults: Findings from The Irish Longitudinal Study on Ageing (TILDA). *Experimental Gerontology*, 183:112314, 2023. ISSN 0531-5565. doi: 10.1016/j.exger.2023.112314. 15

[38] Zahra Azizi, Rebecca J. Hirst, Alan O' Dowd, Cathal McCrory, Rose Anne Kenny, Fiona N. Newell, and Annalisa Setti. Evidence for an association between allostatic load and multisensory integration in middle-aged and older adults. *Archives of Gerontology and Geriatrics*, 116:105155, 2024. ISSN 0167-4943. doi: 10.1016/j.archger.2023.105155. 15

[39] Soraya Matthews, Mark Ward, Anne Nolan, Charles Normand, Rose Anne Kenny, and Peter May. Predicting mortality in The Irish Longitudinal Study on Ageing (TILDA): Development of a four-year index and comparison with international measures. *BMC Geriatr*, 22:510, June 2022. ISSN 1471-2318. doi: 10.1186/s12877-022-03196-z. 16, 18, 19

[40] James Davis, Silvin P. Knight, Rossella Rizzo, Orna A. Donoghue, Rose Anne Kenny, and Roman Romero-Ortuno. A linear regression-based machine learning pipeline for the discovery of clinically relevant correlates of gait speed reserve from multiple physiological systems. In *2021 29th European Signal Processing Conference (EUSIPCO)*, pages 1266–1270, August 2021. doi: 10.23919/EUSIPCO54536.2021.9616187. 17, 19